# Helios Compute Control - illustrative venture investment committee memorandum

Illustrative synthetic case - not investment advice

**Current decision: HOLD - LOSS HURDLE NOT MET.**

## Recommendation and executable terms

Hold because, under the canonical unreviewed synthetic prior, 20.00% of retained paths fall below 1.0x versus the illustrative 10.00% policy maximum. This result is driven by the explicit catastrophe-state assumption; the proposed milestone financing is a path to reconsideration, not a current investment recommendation.

**Binding loss hurdle:** 20.0% versus <= 10.0%; status Misses. The conditional posture is not funding approval while this test and the diligence gates remain open.

| TERM | EXACT SELECTED-CASE POSITION |
|---|---|
| Security | Series C, 1x non-participating preference, senior to A/B and junior to any declared bridge |
| Initial / conditional capital | $25M at close / $15M at month 12 |
| Valuation | $160M pre-money / $200M fully funded post-money |
| Series C ownership | 13.51% at first close / 20.00% fully funded |
| Option pool | 1,611,031 unissued shares after refresh; pre-money holders bear the refresh |
| Selected milestone returns | 56.3% gross XIRR / 8.2x gross MOIC |
| Down-round returns | 7.3% gross XIRR / 1.4x gross MOIC |
| Shortfall case | Bridge at month 36; 17.8% / 2.3x |

## Product, market, customers, competition, and business model

Helios can become the system of control for volatile enterprise GPU spend if ordinary cohorts retain and optimizer savings translate into durable platform economics.

| DIMENSION | UNDERWRITING JUDGMENT | DISCONFIRMING FACT / OPEN GATE |
|---|---|---|
| Product | GPU-spend control plane with customer-level usage and cost records. | Production replication of the optimizer effect remains open. |
| Market | Tiered Bayesian survey supports a scenario range, not a market fact. | Tier 5 is data-thin and abstained; priors remain explicit. |
| Customers | Ordinary cohorts are separated from hand-picked design partners. | Pooled retention overstates repeatability. |
| Competition | Workflow value depends on neutral provider-cost control. | Provider-native tooling and cloud concentration may compress differentiation. |
| Business model | Usage-linked expansion can compound with customer compute spend. | Gross margin remains exposed to compute, telemetry, and support costs. |
| Terms | Milestone financing limits capital at risk before operating proof. | Failure creates explicit down-round or bridge dilution. |

**Counterthesis:** Design-partner selection, inflated pipeline, cloud-cost exposure, and preference-heavy outcomes may make growth and TAM appear more durable than they are.

## Team and financing accountability

These are role-level evidence states, not invented biographies or reference-checked conclusions.

| DIMENSION | ASSESSMENT |
|---|---|
| Observable strengths | CEO / founder: Product insight is supported by design-partner problem definition. CTO: Optimizer experiment shows synthetic unit-cost improvement. CRO / revenue: Pipeline stage history exists at opportunity level. Finance: No dedicated capability is evidenced. |
| Unproven capabilities | CEO / founder: Commercial dependence and succession remain open. Evidence state: OPEN. CTO: Provider concentration and production replication remain open. Evidence state: PARTIAL. CRO / revenue: Ordinary-customer repeatability is not established. Evidence state: OPEN. Finance: Usage margin, runway, and financing controls need an accountable owner. Evidence state: ABSENT. |
| Key-person risk | OPEN - founder commercial dependence and succession evidence remain conditions to the second tranche. |
| Required capacity | Finance leader accountable for usage margin, monthly close, runway, and financing controls. Revenue operations owner accountable for ordinary-customer stage-to-close governance. |

## Cap table and financing-event bridge

| SCENARIO | FUNDED SERIES C | OWNERSHIP | MINIMUM CASH | EXIT PROCEEDS | GROSS XIRR | GROSS MOIC |
|---|---|---|---|---|---|---|
| Base | $25M | 12.94% | $13.4M | $160.3M | 45.0% | 6.4x |
| Milestone-funded | $40M | 20.00% | $17M | $326.9M | 56.3% | 8.2x |
| Downside | $25M | 10.14% | $9M | $35.6M | 7.3% | 1.4x |
| Financing shortfall | $25M | 10.79% | $250,000 | $56.7M | 17.8% | 2.3x |

| SELECTED EVENT | MONTH | STATUS | CASH | NEW SHARES | FULLY DILUTED SHARES |
|---|---|---|---|---|---|
| series-c-close | 1 | FUNDED | $25M | 1,814,223 | 13,425,254 |
| series-c-tranche | 12 | FUNDED | $15M | 1,088,534 | 14,513,788 |

## Milestone financing and monthly runway

The $15M second tranche releases only if every test below passes after a 30-day cure. The board finance committee evaluates the evidence. If any test fails, the tranche is withheld and runway is re-underwritten.

- Ordinary-customer NRR is at least 105.0% across the trailing twelve months.
- Gross margin is at least 70.0% across the trailing three months.
- The opportunity-stage history is complete as of the test date.
- The optimizer result has been replicated before the second tranche.

| YEAR | ENDING CASH |
|---|---|
| 1 | $48.2M |
| 2 | $38.5M |
| 3 | $30.2M |
| 4 | $23M |
| 5 | $17M |

## Preference waterfall and investor return bridge

The selected exit equity value is not a naked outcome assumption. It is derived from the retained LTM revenue ledger and a declared five-year operating and valuation scenario:

| EXIT-VALUE OPERAND | SELECTED MILESTONE CASE | CLASSIFICATION |
|---|---|---|
| Observed LTM revenue | $57M | Retained synthetic P&L |
| Annual revenue growth / hold | 48.0% / 5 years | Scenario |
| Terminal revenue / exit multiple | $404.4M / 4.0x | Scenario |
| Exit enterprise value / cash | $1,617.6M / $17M | Accounting bridge; net debt is -$17M |
| Exit equity value | $1,634.6M | Waterfall operand |

Exit value basis is `Equity value`. Unissued option-pool shares are treated as fully granted common at exit and share pro rata in residual proceeds. Class proceeds conserve to $1,634.6M with a 0-cent residual.

| CLASS | ELECTION | PREFERENCE | RESIDUAL | TOTAL PROCEEDS |
|---|---|---|---|---|
| Series A | CONVERT | $0 | $230.9M | $230.9M |
| Series B | CONVERT | $0 | $208.4M | $208.4M |
| Series C | CONVERT | $0 | $326.9M | $326.9M |

Series C invests $40M in dated funded tranches and receives $326.9M at exit: **56.3% gross XIRR / 8.2x gross MOIC**.

## Conditional distribution and sensitivity

The 1,000 retained paths replay the full financing ledger and exact waterfall. They are scenario priors, not a forecast or evidence of real-world accuracy.

Declared scenario-state priors: Base 30.0%; Downside 15.0%; Financing Shortfall 10.0%; Milestone 45.0%.

| STATISTIC | P10 | P50 | P90 |
|---|---|---|---|
| Gross MOIC | 0.0x | 5.1x | 9.9x |
| Gross XIRR | -100.0% | 36.4% | 61.8% |

- Probability below 1.0x: **20.0%** across 1,000 retained scenario paths.
- Sensitivity book: 14 full-engine cells across exit value, exit date, later-round price, and milestone state.

### Risk policy and decision sensitivity

Helios growth investment screening profile: maximum probability below 1.0x of 10.0%; state DRAFT.

6 retained risk-sensitivity cases are available for review.

## Value creation and board cadence

Combined full-model impact: **$3M minimum cash**, **$3.4M Series C proceeds**, **0.3% gross XIRR**, including an explicit **$0 interaction residual**.

| INITIATIVE | EVIDENCE CLASS | BASELINE / TARGET | OWNER | MODELED CONSEQUENCE | STOP RULE / RISK |
|---|---|---|---|---|---|
| Ordinary-cohort expansion | Analyst judgment | 117.07% / 125% | CRO | $75,504/month cash; $9.1M exit value; $1.5M cost | Design-partner tactics do not transfer |
| Optimizer unit economics | Synthetic causal analysis plus scenario assumption | 70.61% / 74% | CTO | $53,661/month cash; $7.7M exit value; $2M cost | Provider price changes or workload mix invalidate transferred savings |
| Enterprise sales governance | DESCRIPTIVE | 48 inflated opportunities / <15% forecast error | CRO / Finance | $0/month cash; $0 exit value; $1.2M cost | Enterprise cycle elongation remains unidentified |

## Falsifiers, open diligence, and limitations

- FALSIFIER: Ordinary-cohort NRR below 100.00%
- FALSIFIER: Pipeline conversion below 20.00%
- FALSIFIER: Gross margin below 65.00%
- FALSIFIER: Runway below 12 months post-close
- **Loss hurdle exceeds policy maximum** Owner: Illustrative investment committee. If unresolved: Hold under the canonical synthetic prior. Do not revise the prior or policy after observing the result; alternative values remain unreviewed what-if sensitivities.
- **Reconcile pipeline stage history** Owner: Revenue operations diligence lead. If unresolved: Do not credit forecast conversion until the complete opportunity history is reconciled.
- **Test ordinary-customer transferability** Owner: Commercial diligence lead. If unresolved: Remove pooled-retention credit if design-partner and ordinary-customer terms are not comparable.
- **Reconcile provider unit costs** Owner: Technical and financial diligence leads. If unresolved: Do not release milestone capital until margin improvement ties to invoices and telemetry.
- **Reconcile executed financing terms** Owner: Deal counsel. If unresolved: Do not fund until ownership and every waterfall scenario tie to executed terms.
- **Replicate optimizer economics** Owner: Technical diligence lead. If unresolved: Do not assign base-case value or release contingent capital without replication.

This packet is generated from a fictional deterministic room. Synthetic causal estimates recover planted mechanisms only. Scenario outputs are not forecasts. The investment record remains unsigned and requires human IC authority.